**7** The electric starter motor in a car is switched on and off using a relay.

The relay consists of a plastic case and two flexible springy strips, X and Y, which are made of soft iron. These iron strips act as the switch when a circuit is connected between the terminals W and Z.

Fig. 7.1 shows X, Y and the plastic case.

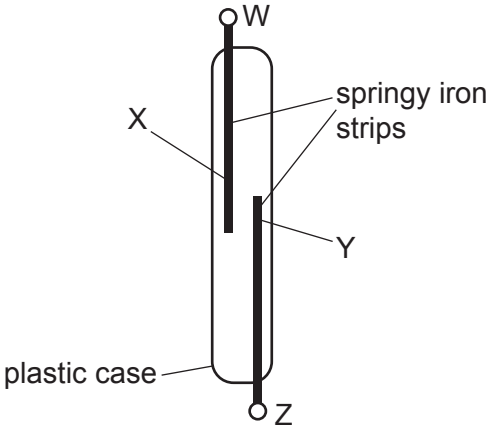



**Fig. 7.1**

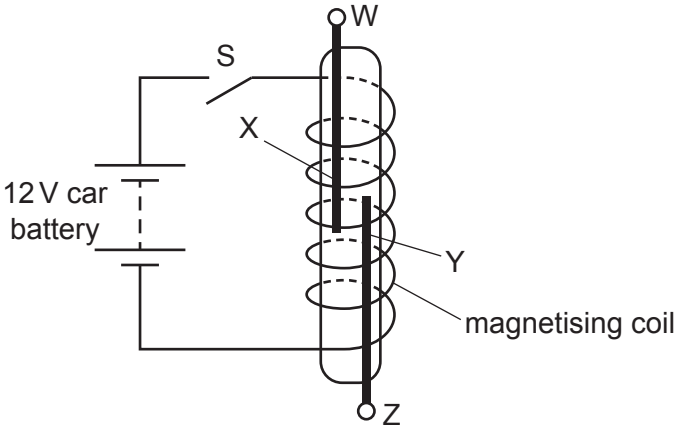



**Fig. 7.2**

Fig. 7.2 shows the equipment from Fig. 7.1 inside a magnetising coil. The magnetising coil is in series with the 12 V car battery and switch S, which is open.

**(a)** Switch S is now closed.

Explain what happens to the springy iron strips X and Y.

.....

.....

.....

..... [3]

**(b)** The power of the starter motor is 1.8 kW and it is also operated by the car battery.

**(i)** Calculate the current in the starter motor when it is used.

current = ..... [2]

**(ii)** The starter motor circuit is connected between terminals W and Z.

Explain why copper wires with a large cross-sectional area are used for this circuit.

.....

.....

..... [2]

**(c)** Fig. 7.3 shows the relay and the symbols for the car battery and the starter motor.

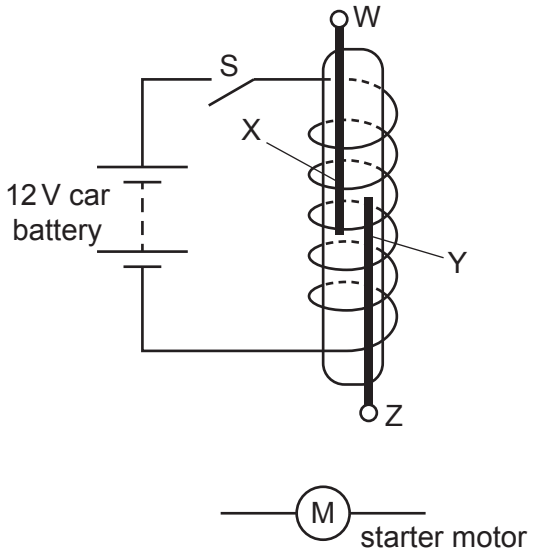



**Fig. 7.3**

The springy iron strips X and Y act as the switch for the starter motor circuit.

Complete the circuit diagram for the motor circuit. [2]

[Total: 9]